

ASHUTOSH MISHRA

SENIOR LLM ENGINEER | GENAI, RAG

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PROFESSIONAL SUMMARY

Senior LLM Engineer and AI Consultant with 8+ years of experience across software, data and AI delivery, including 5+ years building production-grade GenAI, RAG, NLP, multilingual translation and real-time conversational systems. Hands-on expertise in Azure OpenAI, Microsoft Fabric, FastAPI, Python, LangChain, LlamaIndex, Milvus, Docker, Kubernetes and NVIDIA DGX. Experienced in LLM fine-tuning, semantic retrieval, evidence grounding, citation verification, model evaluation, API design, distributed deployment and observability. Known for translating complex business requirements into secure, scalable and auditable AI products for consulting, government and legal use cases.

CORE TECHNOLOGIES

GenAI / LLM: Azure OpenAI, Retrieval-Augmented Generation (RAG), agentic AI, prompt engineering, LLM evaluation, fine-tuning, embeddings, reranking, structured output, hallucination controls, LangChain, LlamaIndex, NVIDIA NeMo, NVIDIA Riva

Cloud / DevOps: Microsoft Azure, Microsoft Fabric, Fabric SQL, Azure AI services, Docker, Kubernetes, Nginx, CI/CD, distributed GPU deployment, container orchestration, autoscaling, secrets/configuration management

Backend / Data: Python, FastAPI, REST APIs, async processing, MongoDB, Milvus, vector databases, Pandas, NumPy, scikit-learn, PDF/HTML/XML processing, metadata and document pipelines

Real-Time / MLOps: LiveKit, WebRTC, OpenAI Whisper, speaker diarization, Microsoft Teams Bot Framework, Prometheus, Grafana, Loki, Locust, Postman, DVC, Git/GitHub, NVIDIA DGX

PROFESSIONAL EXPERIENCE

Deloitte Touche Tohmatsu India LLP | Senior Consultant, Data Science & AI Incubator | Dec 2025 - Present | Gurugram, India

APSC Capability Review Evaluator - Senior LLM Engineer | Azure OpenAI, FastAPI, Microsoft Fabric SQL, Python

Own the end-to-end LLM architecture for an Azure-native evaluator that converts live interview transcripts into structured capability assessments aligned with the Australian Public Service Capability Review Framework.

Built transcript segmentation and semantic classification pipelines for domains and elements, maturity ratings (Emerging, Developing, Embedded, Leading), strengths, gaps, opportunities, risks and evidence-based improvement actions.

Designed high-precision contradiction detection using atomic proposition alignment across subject, predicate, scope, timeframe, population, metric and polarity; excluded merely related, complementary or temporally different statements from contradiction outputs.

Implemented authoritative PDF grounding using full-document processing for short files and page-aware retrieval for long files; returned exact source passages, page references, rationale and Supported/Contradicted/Unvalidated confidence outcomes.

Created stateless, UI-ready FastAPI services with JSON schema validation, retry handling, database connection cleanup, deduplication, evidence traceability and synthetic evaluation suites across 9 Capability Review elements.

Maintained engineering quality through modular repository architecture, code review discipline and a current Pylint score of 9.55/10.

Agentic AI and Meeting Intelligence | FastAPI, Azure, LiveKit, Whisper, Reflex, Teams Bot

Delivered an agentic knowledge assistant for Bridgestone Thailand, reducing information-retrieval time by an estimated 30%.

Architected Meeting Room, a reusable full-stack meeting platform with FastAPI and Reflex; integrated LiveKit WebRTC for low-latency audio/video and Whisper for timestamped transcription and speaker diarization.

Built a Microsoft Teams interview assistant that joins live calls, separates interviewer/candidate dialogue, classifies answers against configurable skills and domains, generates context-aware follow-up questions and produces post-interview reports.

Worked across product, UI, data and assessment stakeholders to define API contracts, review model behaviour, explain confidence and evidence limitations, and turn prototype workflows into maintainable project assets.

Applied Azure-native design patterns for configuration, secure service integration, monitoring and controlled failure handling while keeping user-facing outputs concise, explainable and audit-ready.

National Informatics Centre (NIC) | Senior Software Developer, Data Science, AIRD | Jul 2022 - Dec 2025 | New Delhi, India

AI Anveshika - Citation-Critical Legal and Government RAG | FastAPI, Milvus, MongoDB, NVIDIA DGX, Docker, Kubernetes

Architected a legal RAG platform where factual accuracy and correct citations were mandatory, enabling users to retrieve precedents, case histories and law references with traceable source evidence for e-Courts and Supreme Court workflows.

Built ingestion and retrieval pipelines covering PDF-to-Markdown conversion, document cleaning, chunking, metadata management, NVIDIA embeddings, Milvus vector indexing, semantic search, reranking and citation-aware answer generation.

Reduced information-retrieval time by an estimated 30% for e-Courts and 40% for Ministry of Finance use cases through optimized indexing, retrieval and response generation.

Containerized model and API services with Docker and orchestrated workloads on Kubernetes; used Nginx, health checks, resource limits, GPU scheduling and monitoring to support reliable multi-service deployment.

Optimized and served OSS GPT-120B, Mistral-7B and Nemotron-Llama-3.1-253B on distributed NVIDIA DGX infrastructure, balancing latency, throughput, GPU memory and context-window requirements.

AI Panini - Multilingual Transformer Translation | *IndicTrans2, Gemma 3, FastAPI, Docker, Kubernetes*

Fine-tuned an 18x18 Indic-language transformer on custom parallel corpora, improving IndicTrans2 translation accuracy to 88% across government language workflows.

Led a document-level translation engine for 22 official Indian languages using a fine-tuned Gemma 3 4B-IT model with 8K-token context, preserving document meaning across web pages, scientific text and code comments with up to 92% accuracy.

Developed production APIs for PDF, DOC, HTML and XML translation, plus a Chrome extension for full-page multilingual translation; packaged services in Docker and deployed scalable workloads through Kubernetes.

Secure Model Serving and MLOps | *Docker, Kubernetes, NVIDIA DGX, Prometheus, Grafana, Loki*

Built containerized inference stacks for LLM, embedding and speech services, including environment isolation, persistent model storage, GPU-aware scheduling, horizontal scaling and API health monitoring.

Implemented load and reliability testing with Locust and Postman, operational dashboards with Prometheus/Grafana, centralized logs through Loki and versioned experimentation with Git and DVC.

Antero Technology Group | Data Analyst | *Jul 2019 - Jun 2022* | *India*

Delivered data collection, cleansing, exploratory analysis, visualization, statistical modelling, forecasting and client reporting across business engagements using Python, Pandas, NumPy and scikit-learn.

Translated stakeholder questions into measurable analytical outputs, automated recurring data preparation tasks and presented findings through clear reports and decision-focused visualizations.

Signature Enterprises and Other Real-Estate Firms | Operations, Site Supervision and Sales | *2011 - 2017* | *India*

Managed site operations, vendor coordination, progress tracking and customer-facing sales activities, building early experience in delivery ownership, stakeholder communication and problem solving.

SELECTED TECHNICAL ACHIEVEMENTS

Grounded AI: Designed verifiable evidence workflows with exact citations, confidence scoring, source-page traceability, structured JSON validation and safe fallbacks for missing or conflicting evidence.

LLM Quality: Built curated evaluation datasets for genuine contradictions, semantic non-contradictions, scope/time mismatches, duplicate evidence, long-document retrieval and multilingual edge cases.

Azure Delivery: Integrated Azure OpenAI and Microsoft Fabric SQL into production-style FastAPI services with configurable prompts, controlled retries, secure configuration and auditable processing.

Cloud-Native Engineering: Productionized multi-container AI services with Docker and Kubernetes, including GPU workloads, service isolation, scaling, monitoring, logging and deployment automation.

EDUCATION

M.Tech., Artificial Intelligence & Machine Learning | BITS Pilani | 2023 - 2025

PG Diploma, Artificial Intelligence | CDAC Noida | 2021

M.Sc., Computer Science | Himalayan Garhwal University | 2017 - 2019

BCA, Computer Applications | CCS University | 2008 - 2011